

**Question:**

An experimental infusable drug, X729, is currently being studied to determine its pharmacokinetics. The drug was found to have a half life of 1.5 hours and is eliminated by first order kinetics. What is the minimum number of hours required to reach a steady state concentration of >90%? Answer Choices: A. 1.5 B. 4.5 C. 6 D. 7.5

**Reasoning + Answer:**

<think> To approach this reasoning problem, we need to analyze the context of reaching a steady-state concentration in the context of pharmacokinetics, specifically focusing on first-order kinetics and half-life. Given that the half-life is 1.5 hours, which is a common measure in such scenarios, we can deduce that reaching a concentration greater than 90% would require approximately 4.5 hours, as this is the time it takes for a substance to reach a steady state in a first-order process. This calculation is based on the understanding that after four half-lives, the concentration approaches 97.5% of the final value, which is effectively more than 90%.</think><answer> 4.5 </answer>

